



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology Academic Year 2024 - 2025 (Odd Semester)

Semester, Degree & Branch: V Semester, B.Tech. IT

Course Code & Title: CS3551 & Distributed System

Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Synchronous versus Asynchronous Executions, Design Issues and Challenges
- **Course Outcome:** The students will be able to elucidate the foundations and design issues of distributed computing.
- **Activity Chosen:** Learning by Teaching

- **Justification:**

In unit I, the students need to understand the concepts of Synchronous versus Asynchronous Executions and Design Issues and Challenges. This activity helps the students to understand the design issues and challenges of distributed system.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

At the end of the lecture, students were instructed to write about the fundamentals learned in design thinking process. During the last 20 minutes, A topic was given to Ms. S. Sobana presented. The first Sobana explain the definition for synchronous and Asynchronous execution to 10 minutes. Then remain 10 minutes she continues with Design issues and challenges explain in distributed computing. The implementation of Learning by teaching activity as shown in figure1 and figure2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	1	3	3	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PSO1	PSO2	PSO3
	3	2	1	3	3	1
Justification for correlation	Students could apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.	Students will be able to identify, analyze and define complex problem of distributed computing	Students can elaborate the design issues and challenges in distributed computing	The basic understanding of the designing will enable the students to develop software to solve the complex computation tasks.	Students could apply the knowledge of features supported by various UI design in developing reliable IT solutions.	Students can be able to apply appreciable knowledge of design and development practices to deliver a quality outcome in software project.

• **Images / Screenshot of the practice:**



Figure 1. Learning by Teaching by Sobana S



Figure 2. Learning by Teaching

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- The students felt that they can able to understand of design issues and challenges for distributed computing.
- Students explored the importance of synchronous and asynchronous execution.
- It makes the students to gain knowledge of how to pass the message in processor in distributed system.

❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
- Students can able to explain the topic without any confusion.
- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.

❖ ***Challenges faced in implementation:***

- The presented student hesitated and lack to explain the concepts.
- Student felt that the content is too vast, so that its difficult to complete the presentation on-time.

References:

- <https://scalac.io/synchronous-vs-asynchronous/>
- <https://www.youtube.com/watch?v=EH31Dta6A9c>
- https://medium.com/@ak_gaur/design-issues-in-distributed-system-00ac0ac25ce3
- <https://www.youtube.com/watch?v=Ej3OiG2-6B8>

Signature of Faculty Member

HOD